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ORIGINAL ARTICLE

Aspirin and pre-eclampsia prevention in women with pre-existing diabetes: a retrospective study

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Key words
aspirin, diabetes, pre-eclampsia, postpartum haemorrhage, glycated haemoglobin A.

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Abstract

Background: Aspirin is routinely prescribed in high-risk pregnancies to prevent pre-eclampsia; however, there is a paucity of data in women with pre-existing diabetes.
Aims: To assess the efficacy and safety of aspirin in women with pre-existing diabetes in preventing pre-eclampsia.
Methods: A retrospective review of women with pre-existing diabetes who attended antenatal clinics in a tertiary referral hospital between 2013 and 2019 was conducted. Cases were those receiving aspirin prior to 16 weeks, with pre-eclampsia as the primary outcome. The relationship between early pregnancy glycaemic control and pre-eclampsia was also assessed.
Results: Of the 164 women included in the study, 45 received aspirin. There were no differences in pre-eclampsia (odds ratio (OR) 0.9 (0.3–3.0), $P = 0.924$) or any other measure of placental insufficiency (OR 1.7 (0.7–4.3), $P = 0.243$) between the aspirin and control groups after adjusting for baseline differences. Aspirin therapy was associated with an increased risk of postpartum haemorrhage (PPH) (OR 3.1 (1.1–9.1), $P = 0.041$). The incidence of pre-eclampsia increased stepwise according to early pregnancy HbA1c subgroups of ≤6.0% ($n = 47$), 6.1–7.5% ($n = 57$) and > 7.5% ($n = 39$), with rates of 0, 12.3 and 20.5% ($P = 0.007$) respectively.
Conclusions: The aspirin group had a higher baseline risk of pre-eclampsia and placental insufficiency, therefore the absence of difference between the groups favoured the efficacy of aspirin. PPH was highlighted as a potential complication of therapy, and early pregnancy HbA1c as a novel risk stratification tool for pre-eclampsia in women with pre-existing diabetes.

Introduction

Pre-eclampsia is a multi-system obstetric disorder that is an important cause of morbidity and mortality for the mother and baby. Despite extensive scientific efforts to understand the aetiological aspects of pre-eclampsia, prevention of the disease has been historically challenging, with its prevalence remaining relatively unchanged at about 3–5% despite advances in perinatal care.[1,2] The prevalence among women with pre-existing diabetes is even higher, estimated at 9–24%.[3]

In the past 30 years, aspirin has been extensively investigated as a potential agent to prevent pre-eclampsia. Aspirin inhibits the biosynthesis of thromboxane-A2, a key mediator of the vasospasm and coagulation abnormalities that precede pre-eclampsia.[4] The use of aspirin in obstetric care commenced after a randomised study of 102 women in 1985 suggested a protective effect.[5] In 2007, a meta-analysis of individual patient data was performed, synthesising data from over 30 randomised controlled trials involving 32 217 women with risk factors for pre-eclampsia.[6] They demonstrated a 10% reduced pre-eclampsia risk in women given aspirin, but were unable to identify any particular subgroups of women most likely to benefit. Identification of suitable candidates for aspirin remains an area of controversy.

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In particular, there has been a paucity of data examining the efficacy and safety of aspirin specifically in the high-risk group of women with diabetes. In 2000, a prospective study of 462 women with pre-existing diabetes comparing aspirin to placebo revealed no difference in the rate of pre-eclampsia.[7] However, this study only included women with type 1 diabetes and lacked information on glycaemic control. Furthermore, safety issues, such as postpartum haemorrhage (PPH), were not investigated, and it is well recognised that women with diabetes may be particularly susceptible to this complication.[8]

Considering the paucity of data, our retrospective review aims to assess the efficacy and safety of aspirin in women with pre-existing type 1 or type 2 diabetes in preventing pre-eclampsia.

Methods

A retrospective review of women with pre-existing type 1 or type 2 diabetes delivering between January 2013 and February 2019 was conducted at a tertiary referral hospital. Women with pregnancy loss or termination prior to 20 weeks were not included in the database. Women with gestational diabetes, overt diabetes in pregnancy, multiple pregnancies or incomplete follow up were excluded. The study was approved by the Western Sydney Local Health District Research Ethics Committee (4037 – LNR/14/WMEAD/236 – LNRSSA/14/WMEAD/262).

Women with pre-existing diabetes were managed in a multidisciplinary Diabetes in Pregnancy clinic, where they were reviewed by the obstetric and endocrinology teams, and for patients with hypertension or diabetic nephropathy, by the renal team. The decision to prescribe aspirin was made by any of the teams on the basis of clinical risk of pre-eclampsia. In 2017, a protocol was developed whereby for women at highest risk, aspirin is prescribed at a dose of 100–150 mg daily, generally recommended to be taken at night, aiming to commence prior to 16 weeks gestation and routinely ceased at 36 weeks. All women with pre-existing diabetes, irrespective of aspirin therapy, were reviewed in the Diabetes in Pregnancy clinic every 1–4 weeks, with increasing frequency as the pregnancy progressed. Blood pressure, weight, urinalysis and blood glucose measurements were recorded at each visit in addition to routine antenatal care. Oral antihyperglycaemics were routinely ceased, transitioning to insulin-based therapy alone, and HbA1c and microvascular complications were screened for during pregnancy.

After identifying eligible women, information on aspirin use, age, ethnicity, pre-pregnancy body mass index (BMI), ethnicity-specific BMI categories as per World Health Organization criteria,[9] parity, previous pre-eclampsia, miscarriage, stillbirth, diabetes type, duration from initial diabetes diagnosis, pre-pregnancy diabetes treatment, early pregnancy HbA1c (defined as HbA1c prior to 16 weeks gestation, or if this was not available, preconception HbA1c within 6 months of conception), nephropathy (defined as previous history of nephropathy, or new onset microalbuminuria or elevated creatinine >76 micromol/L prior to 20 weeks gestation[10]), retinopathy, hypertension, smoking and antenatal anticoagulation use were obtained from patient records.

The primary outcome was documentation of pre-eclampsia, which was routinely defined as new onset hypertension after 20 weeks (≥140 mmHg systolic or ≥90 mmHg diastolic) and proteinuria or other maternal organ dysfunction (acute kidney injury, liver derangement, neurological complications, haematological complications, uteroplacental complications), as per the International Society for the Study of Hypertension in Pregnancy (ISSHP) 2014 criteria.[11] In the setting of pre-existing chronic hypertension or pre-existing proteinuria prior to 20 weeks, *de novo* onset of one or more of the above complications with or without exacerbation of hypertension or proteinuria were diagnostic of pre-eclampsia, again in line with ISSHP guidelines. Secondary outcomes included other measures of placental insufficiency (such as small for gestational age (SGA) (<10th percentile) and preterm delivery (<37 weeks)), gestational hypertension, induction of labour, mode of delivery and several neonatal outcomes (neonatal hypoglycaemia, jaundice and respiratory distress). PPH was the primary measure of complications of aspirin therapy, defined as estimated blood loss at delivery exceeding 500 mL. Birthweight centiles were calculated using the GROW Perinatal Institute Bulk Centile Calculator version 8.0.42019,[12] which calculated centiles by correcting for ethnicity, maternal height and weight, parity, gestational age and sex.

The relationship between glycaemic control and the development of pre-eclampsia was examined post-hoc. The incidence of pre-eclampsia was compared in HbA1c subgroups of ≤6.0%, 6.1–7.4% and ≥7.5%. These cut-offs were chosen on the basis of their practicality in the clinical setting, as well as aiming to achieve group sizes as similar as possible.

Statistical analysis

Comparisons between the aspirin and control groups were performed using Pearson χ^2 or Fisher exact tests, and student's *t*-tests for categorical and continuous variables respectively. Categorical variables were summarised by frequencies and percentages, and continuous variables were summarised by mean ± SD if normally distributed

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or median and interquartile range if not normally distributed. Logistic regression was used to estimate the odds of pre-eclampsia and other outcomes in women treated with aspirin and those without aspirin, both before and after adjustment for potential confounding variables. Two-tailed tests with a significance level of 5% were used throughout the study. Analysis was performed using SPSS v25 (IBM, Armonk, NY, USA).

Results

A total of 209 pregnancies met the inclusion criteria, with delivery dates from January 2013 to February 2019. After exclusion for insufficient data on antenatal aspirin therapy or pregnancy outcome ($n = 45$), 164 pregnancies were included for analysis. Of these 164 pregnancies, 45 were deemed high risk by the treating clinician and prescribed aspirin prior to 16 weeks, leaving 119 pregnancies as controls.

Baseline characteristics of those who did and did not receive aspirin are displayed in Table 1. There were no differences between groups in age, ethnic distribution, hypertension and obstetric history. The aspirin-treated group had more background nephropathy ($P = 0.038$), more anticoagulant use ($P = 0.001$), more women in higher BMI categories ($P = 0.038$) and higher early pregnancy HbA1c than controls ($P = 0.038$), however 21 women in the control group did not have early pregnancy HbA1c available.

Outcomes of pregnancies are summarised in Table 2. There was no difference in our primary outcome of pre-eclampsia (odds ratio (OR) 1.3 (0.4–3.5), $P = 0.667$), which persisted after adjusting for differences in nephropathy, BMI category and requirement for anticoagulation using multivariable logistic regression (OR 0.9 (0.3–3.0), $P = 0.924$) (Fig. 1). There were also no differences in SGA and preterm delivery, our secondary measures of placental insufficiency.

We subsequently applied a similar model in the 143 women with HbA1c data available, adjusting for differences in HbA1c in addition to nephropathy, BMI and requirement for anticoagulation, and again there was no difference in pre-eclampsia between the groups (OR 0.8 (0.2–3.1), $P = 0.773$).

The incidence of PPH was significantly greater in the aspirin-treated group (OR 2.7 (1.0–7.2), $P = 0.038$) (Table 2); however, there was more anticoagulant use, higher BMI categories, greater intervention in delivery (defined as Caesarean section or instrumental delivery) and less induction of labour in this group. After adjusting for these differences through multivariable logistic regression, the incidence of PPH remained significantly elevated in the aspirin-treated group (OR 3.1 (1.1–9.1), $P = 0.041$). There were no differences in birthweight or gestation at delivery.

The relationship between glycaemic control and the development of pre-eclampsia was examined, excluding 21 women without early pregnancy HbA1c available, by dividing the cohort into HbA1c subgroups of ≤6.0% ($n = 47$), 6.1–7.5% ($n = 57$) and > 7.5% ($n = 39$). The incidence of pre-eclampsia increased significantly from the lowest to highest HbA1c group respectively (0%, 12.3%, 20.5%, $P = 0.007$) (Fig. 2).

As expected women with early pregnancy HbA1c >6% had significantly more aspirin use ($P = 0.009$), microvascular complications ($P = 0.017$), insulin treatment ($P = 0.008$) and a longer duration from initial diabetes diagnosis ($P = 0.009$). The ethnic distribution also differed significantly ($P = 0.037$), and there was higher incidence of smoking ($P = 0.031$) (Supporting Information Table S1). No women in the lower HbA1c group developed pre-eclampsia, compared to 15.6 of women in the higher HbA1c group (Table S2). In addition HbA1c >6% was associated with other adverse pregnancy outcomes including greater intervention in delivery (OR 2.4 (1.1–4.9), $P = 0.019$) and large for gestational age (LGA) babies (OR 3.6 (1.4–9.3), $P = 0.009$).

Discussion

Our retrospective cohort study compared the pregnancy outcomes of women with pre-existing diabetes who received aspirin prior to 16 weeks to those who did not receive aspirin or received aspirin past 16 weeks. A cut-off of 16 weeks was chosen based on evidence demonstrating no benefit past 16 weeks.[13] Since aspirin was prescribed on the basis of clinical risk, the aspirin group had a higher baseline risk for pre-eclampsia than controls. Thus, the findings need to be considered in the context of this fundamental difference.

We found no difference in the incidence of pre-eclampsia in the aspirin group compared to controls. While this result would typically suggest that aspirin was not effective in preventing pre-eclampsia, the aspirin group had a higher baseline risk, including a significantly higher early pregnancy HbA1c, higher proportion of women with nephropathy, and the presence of women requiring anticoagulation for increased thromboembolic risk or previous evidence of placental insufficiency. Studies have consistently demonstrated that elevations in first trimester and recent preconception HbA1c predispose to pre-eclampsia.[14–16] Diabetic nephropathy is a particularly potent risk factor for pre-eclampsia, with reported rates of 35–64%, compared with 9–17% in women without nephropathy.[17] Maternal thrombophilias and previous placental insufficiency are also associated with

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Table 1 Characteristics of women who received aspirin prior to 16 weeks gestation compared with those who did not

	Aspirin (n = 45)	No aspirin (n = 119)	P-value
Age ± SD (years)	33.2 ± 5.4	32.3 ± 5.7	0.436
Ethnicity			0.736
Caucasian	7 (15.6)	23 (19.3)	
South Asian	16 (35.6)	43 (36.1)	
Other Asian	4 (8.9)	16 (13.4)	
Middle Eastern	11 (24.4)	23 (19.3)	
Aboriginal Australian	1 (2.2)	5 (4.2)	
Pacific Islander	6 (13.3)	9 (7.6)	
Body mass index	30.6 (26.5–32.9)	29.1 (25.6–33.5)	0.153
Body mass index category by ethnicity			0.038*
Underweight	1 (2.2)	0	
Normal	1 (2.2)	19 (16.0)	
Overweight	19 (42.2)	40 (33.6)	
Obese	24 (53.3)	60 (50.4)	
Nulliparous	10 (22.2)	42 (35.3)	0.108
Previous pre-eclampsia†	5 (14.3)	8 (10.4)	0.540
Previous pregnancy induced hypertension†	7 (20.0)	14 (18.2)	0.819
Previous macrosomia†	6 (17.1)	21 (27.3)	0.245
Previous stillbirth†	2 (5.7)	2 (2.6)	0.410
Previous miscarriage‡	23 (54.8)	40 (44.4)	0.269
Diabetes type			0.244
Type 1	4 (8.9)	19 (16.0)	
Type 2	41 (91.1)	100 (84.0)	
Duration from initial diabetes diagnosis (years)	4 (2–9)	3 (1–6)	0.077
Early pregnancy HbA1c (%) (mmol/mol)§	6.8 (6.2–8.2)	6.5 (5.7–7.4)	0.038*
Early pregnancy	41 (91.1)	96 (98.0)	
Preconception	4 (8.9)	2 (2.0)	
Pre-pregnancy diabetes treatment			0.133
No treatment	4 (8.9)	11 (9.2)	
Diet only	6 (13.3)	16 (13.4)	
Oral antihyperglycaemic without insulin	24 (53.3)	60 (50.4)	
Insulin	11 (24.4)	32 (26.9)	
Nephropathy¶	13 (28.9)	16 (13.4)	0.021*
Known nephropathy	10 (22.2)	5 (4.2)	
Early pregnancy microalbuminuria	2 (4.4)	11 (9.2)	
Early pregnancy creatinine > 76 micromol/L	1 (2.2)	0	
Any microvascular complication	15 (33.3)	23 (19.3)	0.058
Smoking	7 (15.6)	13 (10.9)	0.419
Hypertension	6 (13.3)	18 (15.1)	0.772
Anticoagulation during pregnancy††	6 (13.3)	0	0.001*

Expressed as n (%) for categorical, means ± SD for normally distributed continuous variables, median (Q1–Q3) for skewed continuous variables.
*$P < 0.05$.
†Percentages based on n = 112 with a previous pregnancy of at least 20 weeks, with n = 35 for case subjects and n = 77 for control subjects.
‡Percentages based on n = 132 with a previous pregnancy of any duration, with n = 42 for case subjects and n = 90 for control subjects.
§Early pregnancy HbA1c defined HbA1c prior to 16 weeks if available, otherwise preconception HbA1c if within 6 months of conception. The timing of HbA1c measurement in each group is displayed under the mean HbA1c values. Numbers in no aspirin group based on n = 98 as 21 women had no HbA1c data available.
¶Nephropathy defined as previously known diabetic nephropathy, new onset microalbuminuria or newly elevated creatinine >76 μmol/L prior to 20 weeks gestation.
††For the six women on anticoagulation, indications were systemic lupus erythematosus, anti-Ro positivity, previous pulmonary embolism, methylenetetrahydrofolate reductase homozygosity, previous intra-uterine growth restriction and recurrent miscarriage.
SD, standard deviation.

increased pre-eclampsia.[18] Taken in context, our result of no difference in pre-eclampsia between the groups thus favours aspirin as a preventative therapy, as the incidence in the higher risk group appears to have been brought down to baseline. While we were not able to adjust for differences in early pregnancy HbA1c due to

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Table 2 Outcomes of pregnancy comparing aspirin group to controls

	Aspirin (n = 45)	No aspirin (n = 119)	Univariable OR	P-value	Multivariable OR	P-value
Pre-eclampsia	6 (13.3)	13 (10.9)	1.3 (0.4–3.5)	0.667	0.9 (0.3–3.0)†	0.924
Early onset pre-eclampsia	1 (2.2)	5 (4.2)	0.5 (0.1–4.6)	1.000		
Composite placental insufficiency‡	13 (28.9)	25 (21.0)	2.0 (0.8–4.8)	0.286	1.7 (0.7–4.3)†	0.243
Gestational hypertension	2 (4.4)	9 (7.6)	0.6 (0.1–2.7)	0.729		
Pregnancy induced hypertension	8 (17.8)	22 (18.5)	1.0 (0.4–2.3)	0.916		
Antepartum haemorrhage	0 (0)	4 (3.4)		0.576		
Postpartum haemorrhage	9 (20.0)	10 (8.4)	2.7 (1.0–7.2)	0.038*	3.1 (1.1–9.1)§	0.041
Induction of labour	13 (28.9)	66 (55.5)	0.3 (0.2–0.7)	0.002*		
Mode of delivery				0.007*		
Vaginal	11 (24.4)	50 (42.0)				
Elective caesarean	20 (44.4)	22 (18.5)				
Emergency caesarean	12 (26.7)	37 (31.1)				
Instrumental	2 (4.4)	10 (8.4)				
Intervention in delivery¶	34 (75.6)	69 (58.0)	2.2 (1.0–4.8)	0.038*		
Gestation at delivery	38.4 (37.7–38.7)	38.3 (37.4–39.0)		0.369		
Preterm delivery (<37 weeks)	9 (20.0)	21 (17.6)	1.2 (0.5–2.8)	0.728		
Very preterm delivery (<32 weeks)	2 (4.4)	2 (1.7)	2.7 (0.4–20.0)	0.303		
Birthweight centile	75.1 (36.4–91.6)	64.1 (28.4–91.7)		0.391		
Small for gestational age (<10th percentile)	4 (8.9)	5 (4.2)	2.2 (0.6–8.7)	0.260		
Large for gestational age (>90th percentile)	12 (26.7)	34 (28.6)	0.9 (0.4–2.0)	0.809		
5 min APGAR < 9	3 (6.7)	12 (10.1)	0.6 (0.2–2.4)	0.762		
Neonatal hypoglycaemia	13 (28.9)	35 (29.4)	1.0 (0.5–2.1)	0.948		
Neonatal jaundice	7 (15.6)	26 (21.8)	0.7 (0.3–1.7)	0.370		
Neonatal respiratory distress	7 (15.6)	18 (15.1)	1.0 (0.4–2.7)	0.946		
NICU admission	11 (24.4)	34 (28.6)	0.8 (0.4–1.8)	0.597		
Stillbirth	0 (0)	0 (0)				
Neonatal death	0 (0)	1 (0.8)		1.000		
Composite neonatal outcome††	28 (62.2)	71 (59.7)	1.1 (0.6–2.3)	0.765		

Expressed as n (%) for categorical, means ± SD for normally distributed continuous variables, median (Q1–Q3) for skewed continuous variables.
*$P < 0.05$.
†Adjusted for differences in nephropathy, body mass index category and requirement for anticoagulation.
‡Any of pre-eclampsia, small for gestational age (<10th percentile) or very preterm (<32 weeks).
§Adjusted for differences in anticoagulant therapy, body mass index category, induction of labour and intervention in delivery.
¶Caesarean section or instrumental delivery.
††Any of very preterm (<32 weeks), small for gestational age (<10th percentile), hypoglycaemia, jaundice, respiratory distress, NICU admission, 5 min APGAR <9 or neonatal death.
NICU, neonatal intensive care unit; OR, odds ratio; SD, standard deviation.

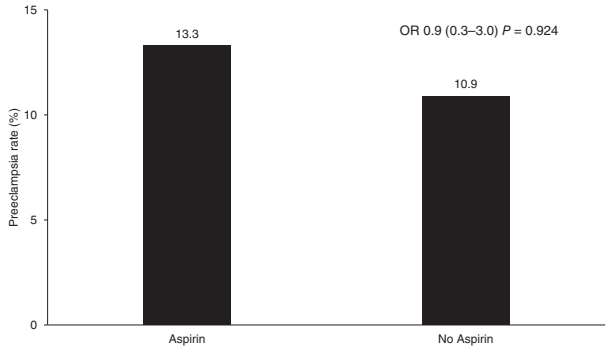


Figure 1 No difference in the incidence of pre-eclampsia between the aspirin and control group. Odds ratio (OR) adjusted for differences in nephropathy, body mass index category and requirement for anticoagulation in pregnancy.

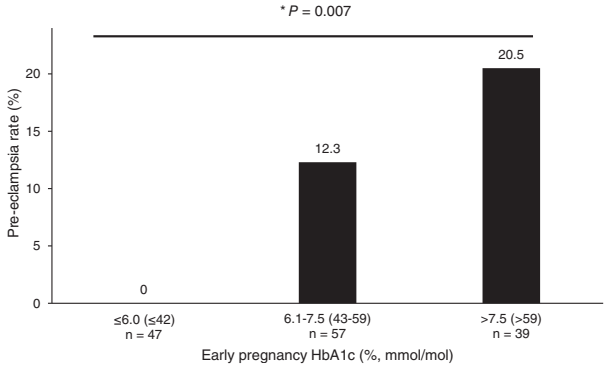


Figure 2 Increasing pre-eclampsia risk with increasing early pregnancy (<16 weeks) HbA1c. P-value significant across groups by Pearson's Chi-squared test. *$P < 0.05$.

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missing data, a multivariable analysis adjusting for differences in nephropathy and women requiring anticoagulation reduced the odds ratio from 1.3 to 0.9, further supporting the efficacy of aspirin. Thus, while we cannot conclude a protective effect of aspirin from our study alone, our data appear to be in line with the global literature that advocates for aspirin as a preventative measure for pre-eclampsia. A Cochrane review published in October 2019[19] demonstrated an 18% risk reduction in favour of aspirin compared to placebo and a number needed to treat of 61, although these figures were not specific to those with pre-existing diabetes. Given the overall incidence of pre-eclampsia in pre-existing diabetes is 9–24%[3] (11.6% in our cohort) compared to the 3–5% observed generally,[1] the benefits of aspirin are likely greater in women with pre-existing diabetes.

The use of aspirin was associated with a significantly elevated risk of PPH. Women prescribed aspirin had a 2.7-fold odds of PPH compared to controls. The increased PPH in the aspirin group is in the context of significantly higher intervention in delivery, 75.6% in the aspirin group compared to 58.0% in controls. However, adjusting for the differences in intervention in delivery, induction of labour and enoxaparin use yielded a further increase in the odds of PPH to 3.1, suggesting that aspirin itself may have contributed to PPH risk. In line with our findings, a recent Cochrane review has linked aspirin to increased PPH risk, although this increase was marginal with a relative risk of 1.09.[19] While our study suggests an association between aspirin and PPH, we cannot decisively comment on the clinical applicability of this finding. Firstly, there are ongoing controversies in the measurement of PPH, with many data suggesting PPH is notoriously underestimated when compared to objective measurements.[20] There is also conflict in establishing a unified definition of PPH. While many institutions have opted to define PPH after Caesarean section as ≥1000 mL to reflect the inherent bleeding propensity of the procedure, there is recent impetus to steer clear of arbitrary numerical cut-offs to define PPH and instead base the diagnosis off early predictors of hypovolaemic shock such as a shock index, calculated by dividing heart rate by systolic blood pressure.[21] Since our study used the traditional definition of estimated blood loss at delivery exceeding 500 mL to define PPH regardless of mode of delivery, and did not include data on haemodynamic changes at delivery, it is unclear whether the increased PPH observed in our aspirin group is of major clinical relevance.

Early pregnancy HbA1c was a strong predictor of pre-eclampsia, with increasing risk with each progression to a higher HbA1c subgroup (Fig. 2), and notably all 47 women with early pregnancy HbA1c ≤6.0% did not develop pre-eclampsia. Elevations in early pregnancy and periconceptual HbA1c may reflect early pregnancy hyperglycaemia that may interfere with placental development and increase pre-eclampsia risk in addition to other adverse pregnancy outcomes.[22] The stepwise increase in pre-eclampsia risk in accordance with HbA1c seen in our study has been demonstrated previously,[15,16] and highlights the potential role of early pregnancy HbA1c in risk stratification of pre-eclampsia. We also showed an increase in intervention in delivery, LGA babies and a potent trend towards adverse neonatal outcomes in our study reiterating the importance of glycaemic control in achieving desirable pregnancy outcomes. In keeping with our study Hughes *et al*. also demonstrated increased adverse outcomes at lower HbA1c levels, with higher incidence of pre-eclampsia, major congenital anomalies, shoulder dystocia and perinatal death in women with booking HbA1c 5.9–6.4% compared to <5.9%.[22] These data suggest advocating for tighter HbA1c targets such as ≤6.0% may be beneficial. Implementing early pregnancy HbA1c into first trimester risk stratification models for pre-eclampsia may be a novel way to assess which women with pre-existing diabetes would benefit most from aspirin when weighing up risks of therapy.

Limitations

In view of the retrospective design, we acknowledge our study has limitations. Notably, the differing risk profiles of the two groups limited our ability to quantify the efficacy of aspirin in women with pre-existing diabetes and the small numbers may have underestimated the difference due to lack of power. Additionally, due to the small sample size we were unable to form separate conclusions for women with type 1 versus type 2 diabetes and acknowledge the findings of our study may be less applicable to women with type 1 diabetes.

Twenty-one women did not have a HbA1c value prior to 16 weeks gestation or within 6 months preconception. However, this was largely due to women presenting late in pregnancy without preconception planning and thus is more likely to underestimate the association with pre-eclampsia. Furthermore, the accuracy of HbA1c reflecting glycaemic control is reduced in pregnancy due to physiological dilution and anaemia.[23] We were unable to examine for iron-deficiency anaemia and other haemoglobin variants such as sickle cell and thalassaemia, which may have affected HbA1c measurements in some of the subjects. Thus, ethnic differences in HbA1c in our study (Table S1) may be related to higher incidence of haemoglobinopathies in some ethnic groups. Nevertheless, HbA1c remains a practical marker of

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glycaemic control in early pregnancy and a novel means of risk stratification for pre-eclampsia among women with diabetes.

The diagnosis of pre-eclampsia in women with pre-existing microalbuminuria/proteinuria, hypertension and/or renal dysfunction remains notoriously difficult. The retrospective design limited our ability to scrutinise the exact criteria used to diagnose pre-eclampsia at the time. Future studies would benefit from clear documentation of the criteria used in order to avoid misdiagnosis. We also acknowledge there was a high prevalence of nephropathy in our cohort despite relatively short duration from initial diabetes diagnosis, suggesting the possibility of non-diabetic kidney disease in these women.

Finally, our study would have benefited from more rigorous assessment of PPH. In addition to estimated blood loss at delivery, further measures such as a shock index, as well as postpartum haemoglobin, requirement for blood transfusion and ICU admission would have helped in distinguishing clinically significant bleeds from those with a high estimated blood loss. Our study highlights that PPH as a potential complication of aspirin therapy should be explored more rigorously in future studies.

Despite these limitations, our retrospective study provides important real world data on the risks versus benefits of aspirin use in women with pre-existing diabetes and highlights the role of early pregnancy HbA1c in risk stratification.

Conclusion

To summarise, although our study design was unable definitively to assess the efficacy of aspirin in preventing pre-eclampsia in women with pre-existing diabetes, our results shed light on important aspects to consider before routinely prescribing aspirin to women with pre-existing diabetes including the role of early pregnancy HbA1c in risk stratification and PPH as a complication of therapy. Randomised prospective studies are required to assess meticulously the benefits and risks of aspirin use, and in particular should aim for rigorous assessment of PPH by clarifying diagnostic cutoffs and documenting peripartum haemodynamic indicators. Overlying these considerations regarding aspirin use is the awareness that preconception glycaemic control is a key preventive strategy for pre-eclampsia and should always be a primary aim prior to conception.

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References

1 Ananth CV, Keyes KM, Wapner RJ. Preeclampsia rates in the United States, 1980-2010: age-period-cohort analysis. *BMJ* 2013; **347**: f6564.
2 Mol BW, Roberts CT, Thangaratinam S, Magee LA, De Groot CJ, Hofmeyr GJ. Preeclampsia. *Lancet* 2016; **387**: 999–1011.
3 Padmanabhan S, Zen M, Lee V, Cheung NW. Pre-existing diabetes in pregnancy. *Minerva Endocrinol* 2016; **41**: 122–37.
4 Sibai B, Dekker G, Kupferminc M. Preeclampsia. *Lancet* 2005; **365**: 785–99.
5 Beaufils M, Donsimoni R, Uzan S, Colau J. Prevention of preeclampsia by early antiplatelet therapy. *Lancet* 1985; **325**: 840–2.
6 Askie LM, Duley L, Henderson-Smart DJ, Stewart LA. Antiplatelet agents for prevention of preeclampsia: a meta-analysis of individual patient data. *Lancet* 2007; **369**: 1791–8.
7 Sibai BM, Caritis S, Hauth J, Lindheimer M, VanDorsten JP, MacPherson C *et al*. Risks of preeclampsia and adverse neonatal outcomes among women with pregestational diabetes mellitus. *Am J Obstet Gynecol* 2000; **182**: 364–9.
8 Kramer MS, Dahhou M, Vallerand D, Liston R, Joseph K. Risk factors for postpartum hemorrhage: can we explain the recent temporal increase? *J Obstet Gynaecol Can* 2011; **33**: 810–9.
9 WHO Expert Consultation. Appropriate body-mass index for Asian populations and its implications for policy and intervention strategies. *Lancet* 2004; **363**: 157–63.
10 Wiles K, Bramham K, Seed PT, Nelson-Piercy C, Lightstone L, Chappell LC. Serum creatinine in pregnancy: a systematic review. *Kidney Int Rep* 2018; **4**: 408–19.
11 Tranquilli A, Dekker G, Magee L, Roberts J, Sibai B, Steyn W *et al*. The classification, diagnosis and management of the hypertensive disorders of pregnancy: a revised statement from the ISSHP. *Pregnancy Hypertens* 2014; **4**: 97–104.
12 Gardosi J, Francis A. Global Bulk Centile Calculator; GROW version 8.0. 4. [cited 2019 Mar 27]. Available from http://www.gestation.net
13 Bujold E, Roberge S, Lacasse Y, Bureau M, Audibert F, Marcoux S *et al*. Prevention of preeclampsia and intrauterine growth restriction with aspirin started in early pregnancy: a meta-analysis. *Obstet Gynecol* 2010; **116**: 402–14.
14 Bashir M, Dabbous Z, Baagar K, Elkhatib F, Ibrahim A, Brich S-A *et al*. Type 2 diabetes mellitus in pregnancy: the impact of maternal weight and early glycaemic control on outcomes. *Eur J Obstet Gynecol Reprod Biol* 2019; **233**: 53–7.
15 Cohen AL, Wenger JB, James-Todd T, Lamparello BM, Halprin E, Serdy S *et al*. The association of circulating angiogenic

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factors and HbA1c with the risk of preeclampsia in women with preexisting diabetes. *Hypertens Pregnancy* 2014; **33**: 81–92.
16 Holmes VA, Young IS, Patterson CC, Pearson DW, Walker JD, Maresh MJ *et al*. Optimal glycemic control, preeclampsia, and gestational hypertension in women with type 1 diabetes in the diabetes and preeclampsia intervention trial. *Diabetes Care* 2011; **34**: 1683–8.
17 Bramham K. Diabetic nephropathy and pregnancy. *Semin Nephrol* 2017; **37**: 362–9.
18 Mello G, Parretti E, Marozio L, Pizzi C, Lojacono A, Frusca T *et al*. Thrombophilia is significantly associated with severe preeclampsia: results of a large-scale, case-controlled study. *Hypertension* 2005; **46**: 1270–4.
19 Duley L, Meher S, Hunter KE, Seidler AL, Askie LM. Antiplatelet agents for preventing pre-eclampsia and its complications. *Cochrane Database Syst Rev* 2019; CD004659.
20 Rath WH. Postpartum hemorrhage–update on problems of definitions and diagnosis. *Acta Obstet Gynecol Scand* 2011; **90**: 421–8.
21 Borovac-Pinheiro A, Pacagnella R, Cecatti J, Miller S, El Ayadi A, Souza J *et al*. Postpartum hemorrhage: new insights for definition and diagnosis. *Am J Obstet Gynecol* 2018; **219**: 162–8.
22 Hughes RC, Moore MP, Gullam JE, Mohamed K, Rowan J. An early pregnancy HbA1c≥ 5.9%(41 mmol/mol) is optimal for detecting diabetes and identifies women at increased risk of adverse pregnancy outcomes. *Diabetes Care* 2014; **37**: 2953–9.
23 Hughes RC, Rowan J, Florkowski CMI. There a role for HbA1c in pregnancy? *Curr Diab Rep* 2016; **16**: 5.

Supporting Information

Additional supporting information may be found in the online version of this article at the publisher's web-site:

Table S1. Characteristics of women with optimal early pregnancy HbA1c compared to those with higher early pregnancy HbA1c.
Table S2. Outcomes of pregnancy comparing women with optimal early pregnancy HbA1c to those with higher early pregnancy HbA1c.

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